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An Overview and Application of Deep Convolutional Neural Networks for Medical Image Segmentation

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***Abstract*— Medical imaging technologies are widely utilized as a non-invasive diagnostic process that helps professionals identify diseases and injuries. Convolutional neural networks (CNNs) have become increasingly popular since they are considered to be the best approach for a variety of computer vision problems includes medical imaging and diagnostics. Image segmentation using CNN models makes image analysis easier that can effectively use with different image modalities like X-ray, MRI, CT, Ultrasound and PET. This review article provides a perspective on using CNN models for various medical image segmentation tasks. It contains the details of benchmark deep CNN models available for image segmentation, publicly available medical imaging datasets and various applications of deep CNN models for medical image segmentation tasks.**

Keywords*—Deep Learning, Convolutional Neural Network, Medical Image Segmentation, U-Net, Fully Convolutional Network*

I. INTRODUCTION

Computer-aided diagnosis greatly assists doctors to interpret medical images with different imaging modalities like X-ray, PET scan, MRI, ultrasound etc. Medical imaging technologies extensively used as a diagnostic non-invasive procedure which allows medical professionals to support in diagnoses diseases and injures. It improves the treatments by allowing doctors to assess human body organs, blood vessels or tissues and bones. It can be used for variety of diagnosis tasks like whether surgery would be required; treatment of bone fractures; assisting doctors in replacement of joints and placement of devices like stents etc. in the body; locating and removal of tumors; identification of blockages or clots in blood vessels and many more.

In recent years, computer-aided diagnosis (CAD) has plays a vital role in medical imaging based diagnosis and treatment. CAD assists radiologist in clinical diagnosis and treatment planning [1]. CAD helps to achieve faster and accurate diagnostic results irrespective to large number of data volume. It even helps and support patient care to remote areas using advanced information technology solutions. Image segmentation in CAD is the process of dividing a single digital image into a number of smaller components known as segments. Regions of Interest (ROI) are extracted from MRI or CT image data as part of the segmentation process for medical pictures. It is a technique that separates a complete image into areas with equivalent texture, contrast and brightness levels, colour, and other attributes [2]. Due to several type of medical image modalities like X-ray, MRI, CT, Ultrasound and PET [3] and varying nature of domain applications like brain tumor detection, skin cancer classification, blood vessels segmentation, it is very complex to devise a generalized segmentation algorithm.

In the field of medical image segmentation, deep learning techniques have recently performed noticeably better than traditional techniques as a result of the availability of more complicated learning algorithms and enhanced hardware capabilities. Convolution Neural Network (CNN) is a deep neural network used extensively in deep learning for the analysis of visual data. Due to its feature extraction capabilities, it is widely used in medical image classification, detection and segmentation [4].

This paper provides a comprehensive review of different deep learning models available for medical image segmentation. The section II describes background of deep learning and CNN. Section III provides a summary of benchmark CNN networks used for image segmentation. Section IV covers the details of several benchmark datasets available for medical image segmentation. Section V discusses the different applications of medical image segmentation with other details like image modality, network used and performance evaluation parameters considered.

II. DEEP LEARNING AND CNN

Deep learning, a part of a larger family of machine learning methods, is built on Artificial Neural Networks (ANN), which are capable of learning representations. Contrary to machine learning, deep learning generates features automatically from raw data rather than requiring that they be explicitly constructed. Deep learning, in contrast to machine learning, does not require manually created features; instead, it automatically creates features from raw data. A multi-layered deep learning architecture called a convolutional neural network (CNN) automatically extracts information from unprocessed input photos. CNN networks are widely employed in many applications [5] due to their sophisticated algorithms, accessibility of enormous amounts of data from multiple sources, and updated hardware facilities. A CNN is a deep neural network-based architecture made up of a stack of layers, each of which performs a different operation, such as convolution, activation, pooling, normalisation, and drop out. Feature extraction and categorization are its two key functions. CNN eliminates the need for human feature extraction by automatically learning features from input data [6]. A collection of pooling layers, numerous convolutional layers, and a set of fully linked layers make up the conventional architecture of CNN. The basic building blocks of a typical CNN are as follows:

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A. Convolution Layer:

A first layer of CNN is a convolution layer that performs convolution operation, a special mathematical operation, for feature extraction [7]. It takes an input matrix of an image and applies a small, normally a size of 3X3 filter on it. The filter is also known as kernel or feature detector. Applying filter on an input matrix at each layer generates feature map that is a representation of features like edges, horizontal and vertical lines, curves etc. To introduce non-linearity, activation functions are used after convolution operation. The most common non-linear operations are ReLU, Sigmoid and Tanh. In past few years, ReLU (Rectified Linear Unit) has become very popular that defined as ReLU(k) = max(0, k) [8]. The feature map generated by convolution layer is feed into an activation function which again produces feature maps that passes to the next layer.

B. Pooling Layer:

Pooling layer is used to the resulting feature map for downsampling, which lowers the dimensionality of feature maps [9]. It aids in lowering the quantity of computation and learning parameters used. The most frequent pooling operations are max, average, and min pooling. Max pooling is the most popular operation and offers superior performance to the other two [10].

C. Batch Normalization and Dropout:

To avoid overfitting in network, batch normalization and dropout are commonly used techniques [11]. Batch normalization normalizes the output of previous layer and used as a regularization that helps overfitting of the model. It applies between layers to speed up training by using higher learning rate to make the learning process easier. Another conventional mechanism used to reduce overfitting in CNN is adding a dropout layer. It simple works by randomly dropping neurons during training in each iteration that slightly modify the network itself for each batch of training data.

D. Fully connected layer:

A CNN network ends with one or more fully connected layers that generates a final outcome. The convolution layer or pooling layer resides at the end of CNN generates an output feature map. This feature map is flattened and connected to a fully connected layer. A fully connected layer is also referred as a dense layer where every input is coupled to every output with a learnable weight [12]. The final completely connected layer has the same number of output nodes as classes. Based on the features acquired from earlier layers, the fully connected layer performs classification.

III. CNN MODELS FOR MEDICAL IMAGE SEGMENTATION

Over the last decade, deep learning models become very popular in computer-aided diagnostic [13]. Moreover, several improvements have been made in CNN architecture to make it more prevailing for prediction tasks [14]. Some of the most influential CNN architectures are the winner of ILSVRC - ImageNet Large Scale Visual Recognition Challenge. This section summarises three prevailing CNN architectures namely Fully Convolution Neural Network, or FCN, Mask-RCNN and U-Net with its network structures that are successfully applied for segmentation medical images with different image modalities.

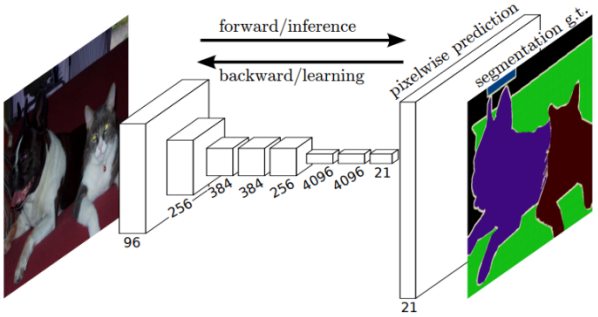


Fig. 1. Fully Convolution Network for Semantic Segmentation [15]

Fully Convolution Neural Network, or FCN, is a form of architecture of CNN which is widely used for semantic segmentation tasks. Unlike to traditional CNNs, FCN does not contains any Dense layers. In FCN, Dense layers are replaced by 1x1 convolutions. The absence of dense layers makes FCN capable to work with variable inputs image dimensions. It contains convolution, pooling and upsampling layers without dense layers and this makes the network faster to train. As represented in Figure 1, with efficient inference and learning technique, an arbitrary size of input is taken by FCN. Moreover, it produces an output of corresponding sized. In order to provide a spatial map that supports image-to-image learning, FCN transforms fully connected layers into convolution layers [15].

Mask-RCNN extends the architecture of Faster R-CNN and offers a general framework for object instance segmentation [16]. Faster R-CNN is a region-based CNN that infers bounding boxes for each object presented in an image along with its class label with confidence score. In addition to detecting objects from an input image, Mask-RCNN generates a segmentation mask for each instance parallel. Mask R-CNN generates masks for each Region Of Interest (ROI) and segments an input image pixel-to-pixel.

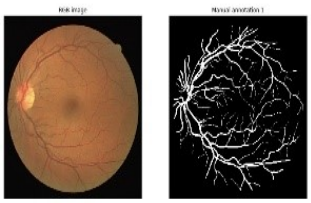
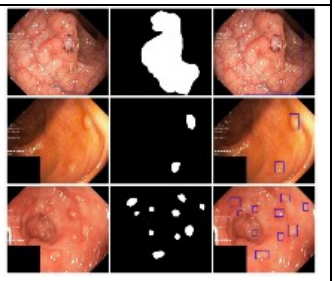
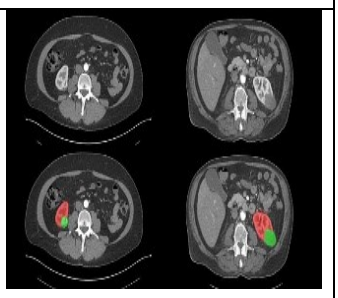
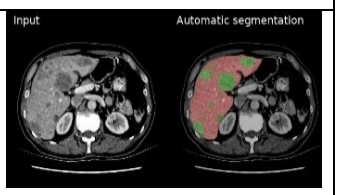
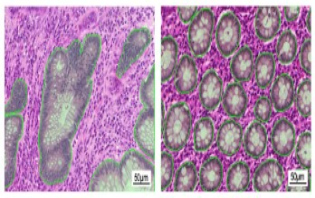
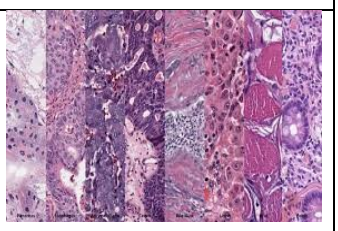
A U-Net, a significantly modified variant of FCN, was created for biomedical image segmentation [17]. It applies skip connections between upsampling and downsampling paths and concatenates instead of adding up the multiple upsampling layers that it comprises. Upsampling path and downsampling path are the two main parts of U-Net. In the downsampling step, convolution blocks and a maxpool downsampling are employed to encode the input image into feature representations at various levels. Semantically developing the discriminative characteristics onto the pixel space is the aim of the upsampling path.

IV. DATASETS UTILIZED FOR MEDICAL IMAGE SEGMENTATION

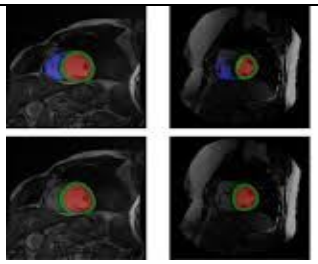
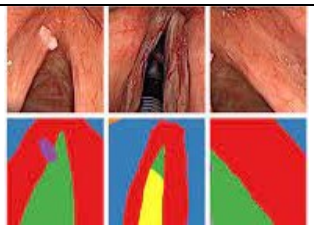
There are various datasets available in public domain that can be utilized for medical image segmentation. The table 1 summarizes the details of different datasets with its applications and usage. It contains the samples of different image modalities like CT, MRI, Microscopic and Ultrasonic images.

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TABLE I. DATASETS FOR MEDICAL IMAGE SEGMENTATION

Name	*Details*	*Samples*	*Sample from Dataset*
DRIVE [18]	Each image in the dataset is having a 584×565 pixel resolution. There are three colour channels each have eight bits.	40 retinal fundus images	
Kvasir-SEG [19]	Images are manually annotated by a medical doctor. An experienced gastroenterologist verified them.	1000 gastrointestinal polyp images	
Medical Segmentation Decathlon [20]	The dataset contains images of different body organ parts like Pancreas, Brain, Prostate, Heart, Hippocampus, Liver, Lung, Spleen, Hepatic Vessel, and Colon	2,633 three-dimensional images	
Liver Tumor Segmentation Challenge 2017 (LiTS17) [21]	The dataset contains the images and its segmentations provided by various clinical sites from all over the world	200 CT scans	
A Challenge for Segmentation of Gland from Colon Histology Images (GlaS) [22]	Zeiss MIRAX MIDI Slide Scanner is used to take images. The pixel resolution is 0.465µm	165 images of histological sections	
PanNuke [23]	The dataset includes labels for the nuclei in 19 different tissue types. This dataset is used for nucleus instance segmentation and classification.	481 visual fields	
ACDC - A Challenge of Diagnosing Cardiac [24]	The dataset is composed of 150 samples that are divided into 5 evenly distributed subgroups	150 MRI Images	

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VocalFolds [25]	This dataset is used to automatically segment laryngeal endoscopic images. Each image has a 512x512 pixel resolution.	536 colour images of the larynx	

V. APPLICATIONS OF DEEP CNNS FOR MEDICAL IMAGE SEGMENTATION

As discussed earlier, deep CNNs are highly adapted by various researchers for recognizing patterns in different input images with diverse image modalities. The selection of specific deep CNN model is depending on image modality, body part(s), segmentation task, disease to be diagnosed etc.

J. Ye *et al.* (2022) explored vanilla U-net for segmenting lesions from whole body FDG-PET and CT scans. They have considered the aspects of data pre-processing and data augmentation along with network architecture. The dice score on the test dataset is 92.81%. G. Chen *et al* (2022) proposed an enhanced selective kernel convolution with U-Net, namely ESKNET for segmenting breast tumor lesions. With convolution layers, different kernel sizes are applied that provides receptive fields with different scales. It improves the network adaptability to deal with different input images. For the evaluation of the network, three publically available ultrasound datasets are considered. The proposed ESKNet improvise the sensitivity and generalization ability.

S. Cerri *et al.* (2022) proposed an open source tool for segmentation of lesions of white matter and whole-brain. They have elaborated a longitudinal method for segmenting longitudinal MRI scans of whole-brain. Authors have validated the proposed method on multiple datasets that contains the samples of control subjects along with the samples of patients having multiple sclerosis and Alzheimer's disease. They have also compared the results of proposed methods against benchmark longitudinal methods.

M. Z. Alom et al. (2018) proposed two new models for medical image segmentations namely RU-Net and R2U-Net. They have conducted experiments considering three diverse datasets. The performance is evaluated against the benchmark segmentation models. RU-Net model is built upon the recurrent convolutional neural network and R2U-Net is based on residual recurrent neural network. Several evaluation metrics are considered for the evaluation includes sensitivity, Dice coefficient, accuracy, specificity, F1-score, and Jaccard similarity. N. V. Keetha et al. (2022) proposed an architecture for lung nodule segmentation and named it as a U-Det. In the proposed architecture, bidirectional feature network is incorporated between encoder and decoder modules. To enhance the segmentation efficiency Mish activation function is used. The experiment is performed on LUNA-16 dataset and obtained 82.82% values for the Dice similarity coefficient (DSC).

T. D. Bui, J. Shin, and T. Moon (2019) proposed a fully convolutional densely connected network or tissue segmentation of 6-month infant brain. They named it as a skip-connected 3D DenseNet. The proposed network provides promising results on iSeg-2017 MRI dataset of infant brain with 91.02 DSC score.

G. Wang *et al.* (2017) proposed to use two CNNs; P-Net and R-Net for placenta segmentation. An initial automatic segmentation is performed by P-Net which is passed to user that provides some interactions to indicate incorrectly segmented regions. These are passed to R-Net for final segmentation. For pancreas segmentation, F. P. Salanitri et al. (2021) proposed a novel fully convolutional network (3D) that segments CT and MRI scans. They have used 3D encoder and 3D decoder to extract features from images at different scans and 4-fold cross-validation is considered for evaluation of the proposed model.

A cascaded fully convolutional neural network is proposed by P. F. Christ *et al.* (2017) for segmentation of CT and MRI images that helpful to quantitative image analysis. In the proposed network, a fully convolutional network is trained first for segmentation of liver images and then the ROI is passed as input to the second fully convolutional network. The dice score achieved by the proposed model is over 94%. A novel method that combines graph convolutional networks with convolutional neural networks is proposed by Z. Zhai et al. (2019) to create classifier for separation of pulmonary artery-vein. The proposed novel model obtained 0.773 validation accuracy that is higher compared 0.817 median accuracy obtained by the observers, and 0.727 obtained by the baseline CNN.

VI. CHALLENGES AND OPPORTUNITIES

There are many challenges observed in the field of computer-aided diagnosis. One of the major concern for researchers is the availability of dataset. It is quite challenging to find adequate medical imaging dataset as

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compare to other domains. The dataset suffers with low resolution images and high level of noise. Moreover, in medical imaging, images are available with different modalities like CT, MRI, PET Scan etc. Once the images are acquired, it should be labelled by an expert medical professional or radiologist.

A process of labelling annotated images involves expert radiologist who identifies area on an images and labeled it. Data annotation is usually a time-consuming task and inaccurate annotation highly impacts the accuracy of the deep learning model.

To train the deep network well, large amount of data is required. Medical images are not openly available; most of the images are contained by diagnostic centers. Also, some of the datasets are suffering from an uneven class distribution problem.

The table 2 summarizes the different applications of deep CNNs with other details like segmentation task, imaging modality, dataset, performance evaluation metrics and implementation details.

TABLE II. APPLICATIONS OF DEEP CNNS FOR MEDICAL IMAGE SEGMENTATION

Segmentation Task	***Modality***	***Organ***	***Dataset***	***CNN architecture***	***Performance Evaluation Metrics***	***Implementation Details***
Lesion Segmentation [26]	FDG-PET/CT	Whole-body	autoPET Challenge	Vanilla U-Net	Dice Score, Fractional Plasma Volume	Pytorch 1.11.0 CUDA 11.2, Tesla A100 80G
Lesion Segmentation [27]	Ultrasound	Breast Tumors	Tumor Dataset	Enhanced selective kernel convolution with U-Net	Jaccard, Precision, Recall, Specificity, Dice Score	TensorFlow-2.6.0, Python 3.6, NVIDIA RTX 3090
Lesion Segmentation [28]	MRI	Whole-brain	AD and MS patient dataset	Longitudinal method	Absolute Symmetrized Percent Change (ASPC)	C++ and Python with FreeSurfer, Intel 12-core i7-8700K
Retinal Vessel Segmentation [29]		Retinal Vessels	DRIVE, STARE, and CHASE_DB	R2U-Net	Sensitivity, Dice Score, Accuracy, Specificity, F1-score, Jaccard	Keras, and TensorFlow, NVIDIA GEFORCE GTX-980, GPU machine with 56GB of RAM
Lung Nodule Segmentation [30]	CT	Lung Nodules	LUNA-16	Modified U-Net	Sensitivity, Positive Predictive Value, Dice Score	Tensorflow2.1, Python3.6, CUDA 10.2, NVIDIA Tesla T4 GPU
Infant Brain segmentation [31]	MRI	Infant Brain	iSeg-2017	Skip-connected 3D DenseNet	Dice Score, Average Surface Distance Modified Hausdorff Distance	Caffe, 12GB Titan X GPU
Placenta Segmentation [32]	MRI	Fetal MRI scans	2D fetal MRI	Deep Interactive Geodesic Framework	Dice Score, Average Symmetric Surface Distance	Caffe, Two 8-core E5-2623v3 Intel Haswells

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Pancreas Segmentation [33]	CT/MRI	Pancreas	NIH Pancreas-CT dataset	3D fully convolutional deep network - PankNet	Dice Score, Positive Predictive Value, Sensitivity	PyTorch and MONAI, NVIDIA Quadro P6000 GPU
Segmentation of Liver and Tumor [34]	CT/MRI	Liver	3DIRCAD	Cascaded FCN (Fully Convolutional Neural Networks)	Dice Score	Titan X GPU (NVIDIA) having 12 GB VRAM
Pulmorary Vessel Segmentation [35]	CT	Lungs	SunYs data set, LUMC data set	CNN-GCN classifier	Accuracy	TensorFlow, Keras 1.12.2, Nvidia Titan Xp 12 GB)

VII. CONCLUSION

In a number of complex tasks of computer vision like image classification, object recognition, and image segmentation, deep learning has emerged as the preferable method. The fundamental ideas behind CNN and deep learning, as well as their implementations, may enhance the effectiveness of computer-aided medical diagnosis. This review article covers the details of applying deep CNN models for medical image segmentation tasks. It provides the fundamentals of deep learning and CNN with benchmark models used for image segmentation. It provides the details of various open datasets available for medical image segmentation with the modality used. Moreover, it summarizes different applications of deep CNNs with other details like segmentation task, imaging modality, dataset, performance evaluation metrics and implementation details.

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